

Question ID: 4b6f0a3f

| Assessment | Test | Domain        | Skill                                                                         | Difficulty                                          |
|------------|------|---------------|-------------------------------------------------------------------------------|-----------------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear equations in one variable and systems of equations in two variables | Medium <div><div></div><div></div><div></div></div> |

Question

$x^2 - 5x - 24 = 0$

What is the sum of the solutions to the given equation?

Correct Answer: 5

Rationale

The correct answer is **5**. The given quadratic equation can be rewritten in factored form as  $(x - 8)(x + 3) = 0$ . Based on the zero product property, it follows that  $x - 8 = 0$  or  $x + 3 = 0$ . Adding **8** to both sides of the equation  $x - 8 = 0$  yields  $x = 8$ . Subtracting **3** from both sides of the equation  $x + 3 = 0$  yields  $x = -3$ . Therefore, the solutions to the given equation are **8** and **-3**. It follows that the sum of the solutions to the given equation is **8 + (-3)**, or **5**.